Parul University NAAC A++ ACCREDITED UNIVERSITY	**Parul University** **Faculty of Engineering & Technology** **Department of Applied Sciences and Humanities** **1st Year B. Tech Programme (CSE (All branches) & IT)** **Semester-2 (2025-26)** **Linear Algebra -(03019102BS01)**

Question Bank-1 (Unit-1: Matrices)

1. Define with an example:

a. Trace of a Matrix

b. Orthogonal Matix

c. Symmetric Matrix

d. Skew Symmetric Matix

e. Diagonal Matix

f. Algebraic Multiplicity of an eigen value

g. Geometric Multipicity of an eigen value

h. Diagonalized Matrix

2. Reduce the following matrices into row echelon form:

(i) $\begin{bmatrix} \mathbf{5} & \mathbf{3} & \mathbf{14} & \mathbf{4} \\ \mathbf{0} & \mathbf{1} & \mathbf{2} & \mathbf{1} \\ \mathbf{1} & \mathbf{-1} & \mathbf{2} & \mathbf{0} \end{bmatrix}$ (ii) $\begin{bmatrix} \mathbf{1} & \mathbf{2} & \mathbf{3} & \mathbf{-1} \\ \mathbf{-2} & \mathbf{-1} & \mathbf{-3} & \mathbf{-1} \\ \mathbf{1} & \mathbf{0} & \mathbf{1} & \mathbf{1} \\ \mathbf{0} & \mathbf{1} & \mathbf{1} & \mathbf{-1} \end{bmatrix}$;

(iii) $\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix}$ (iv) $\begin{bmatrix} 3 & -2 & 0 & -1 & -7 \\ 0 & 2 & 2 & 1 & -5 \\ 1 & -2 & -3 & -2 & 1 \\ 0 & 1 & 2 & 1 & 6 \end{bmatrix}$

3. Find a, b, c, d if $\begin{bmatrix} a & b+c \\ c+d & b \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 3 & -1 \end{bmatrix}$ are equal matrices.

4. Prove that matrix $A = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$ is singular.

5. Is every diagonal matrix symmetric? Justify your answer with appropriate reasons.

6. What is the trace of a diagonal matrix?

7. Reduce the matrix into reduced row echelon form $\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 3 & 6 & 2 \end{bmatrix}$

8. Find trace and determinat value of matix $A=\begin{bmatrix}2&3&4\\4&5&6\\7&8&9\end{bmatrix}$

9. If $\begin{bmatrix}2+x&3&4\\1&-1&2\\x&1&-5\end{bmatrix}$ is a singular matrix, then find the value of x.

10. Find the rank the following matrices by determinant method:

(i) $\begin{bmatrix}2&3&4\\4&3&1\\1&2&4\end{bmatrix}$;(ii) $\begin{bmatrix}4&2&3\\8&4&6\\-2&-1&-\frac{3}{2}\end{bmatrix}$; (iii) $\begin{bmatrix}1&2&3\\2&3&4\\3&5&7\end{bmatrix}$; (iv) $\begin{bmatrix}\mathbf{1}&\mathbf{2}&\mathbf{-1}&\mathbf{-4}\\\mathbf{2}&\mathbf{4}&\mathbf{3}&\mathbf{5}\\\mathbf{-1}&\mathbf{-2}&\mathbf{6}&\mathbf{-7}\end{bmatrix}$

11. Find the rank of the following matrices by Row Echelon Form:

(i) $\begin{bmatrix}3&4&5&6&7\\4&5&6&7&8\\5&6&7&8&9\\10&11&12&13&14\\15&16&17&18&19\end{bmatrix}$ (ii) $\begin{bmatrix}\mathbf{1}&\mathbf{3}&\mathbf{2}&\mathbf{1}\\\mathbf{2}&\mathbf{3}&\mathbf{3}&\mathbf{3}\\\mathbf{3}&\mathbf{4}&\mathbf{-1}&\mathbf{3}\\\mathbf{6}&\mathbf{10}&\mathbf{4}&\mathbf{6}\end{bmatrix}$ (iii) $\begin{bmatrix}\mathbf{3}&\mathbf{2}&\mathbf{1}&\mathbf{4}\\\mathbf{-1}&\mathbf{3}&\mathbf{2}&\mathbf{2}\\\mathbf{2}&\mathbf{3}&\mathbf{3}&\mathbf{6}\\\mathbf{3}&\mathbf{4}&\mathbf{4}&\mathbf{10}\end{bmatrix}$

(iv) $\begin{bmatrix}1&1&-1&1\\1&-1&2&-1\\3&1&0&1\end{bmatrix}$ (v) $\begin{bmatrix}2&4&6\\1&3&2\\3&7&8\end{bmatrix}$ (vi) $\begin{bmatrix}1&0&0\\0&2&4\\1&2&3\end{bmatrix}$ (vii) $\begin{bmatrix}1&2&-3\\3&6&-9\\4&8&-12\end{bmatrix}$

$\begin{bmatrix}1&2&3\\7&8&9\\3&6&9\end{bmatrix}$

12. If the rank of the matrix $A=\begin{bmatrix}1&2&x\\2&4&6\\3&6&9\end{bmatrix}$ is 2 then find value of x.

13. Determine the value of x, for which the matrix $A=\begin{bmatrix}3&x\\2&4\end{bmatrix}$ is singular.

14. If $A=\begin{bmatrix}2&1\\3&4\end{bmatrix}$ and $B=\begin{bmatrix}3&3\\5&2\end{bmatrix}$ then find the value of 2A+3B. Also, if $rank(A)=2$ and $rank(B)=3$ then find the value of $rank(AB)$.

15. What is the rank of the identity matrix I_n?

16. . Solve the following system of linear equations by Gauss- Elimination method

a. x + y + z = 6; x + 2y + 3z = 14; 2x + 4y + 7z = 30

b. $4x + 3y - z = 0$; $3x + 4y + z = 0$; $5x + y - 4z = 0$

c. $x + 2y + 3z = 0$; $2x + 3y + 4z = 0$; $3x + 4y + 5z = 0$

d. $x - y + 2z = 0$; $3x + 4y - z = 0$; $5x + 2y + 3z = 0$

e. $x + 2y + 2z = 0$; $-2x + 5y + 2z = 1$; $8x + y + 4z = -1$

17. . Solve the following system by gauss elimination method

$$\frac{-1}{x}+\frac{3}{y}+\frac{4}{z}=30;\ \frac{3}{x}+\frac{2}{y}-\frac{1}{z}=9;\ \frac{2}{x}-\frac{1}{y}+\frac{2}{z}=10$$

18. . Solve the following system using LU Decomposition Method:

(i) $x + 5y + z = 14;\ 2x + y + 3z = 13;\ 3x + y + 4z = 17$

(ii) $x + y - z = 4;\ x - 2y + 3z = -6;\ 2x + 3y + z = 7$

(iii) $x + y + z = 1$; $3x + y - 3z = 5$; $x - 2y - 5z = 10$

(iv) $x + y + z = 1$; $4x + 3y - z = 6$; $3x + 5y + 3z = 4$

(v) $x + 2y + 3z = 9$; $4x + 5y + 6z = 24$; $3x + y - 2z = 4$

(vi) $x + 6y + 2z = 9$; $2x + 12y + 5z = -4$; $-x - 3y - z = 17$

19. Perform LU Decomposition for the matrix A and find the Lower triangular matrix L and Upper Triangular matrix U where, $A = \begin{bmatrix} 3 & 1 & 6 \\ -6 & 0 & -16 \\ 0 & 8 & -17 \end{bmatrix}$

20. Perform LU Decomposition on the matrix $A = \begin{bmatrix} 4 & 3 \\ 6 & 3 \end{bmatrix}$.

21. If in a 2×2 matrix with trace is 5 and determinant is 6, what is the characteristic equation?

22. If $A = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$ then find the eigen values of A^2 and A^{-1}.

23. Find the product of eigen values of $A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$

24. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ then find eigen values of A^3.

25. Find the characteristic equation of matrix $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$.

26. Find A^5 if $A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$ then find the eigen values for the given matrices (i) A (ii) A^T (iii) A^{-1} (iv) $A^3 + A + I$ (v) $A^3 + 5A + 3I$

27. What is the sum of the eigenvalues of the given matrix $A=\begin{bmatrix}1 & 4\\ 2 & 10\end{bmatrix}$?

28. For a given matrix, $A=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, By using Caley Hamilton theorem, find the value of A^3+A^2-2A

29. Find A^{-1} of the matrix $A=\begin{bmatrix}1 & 2\\ 5 & 6\end{bmatrix}$ by Caley Hamilton theorem.

30. Verify Caley Hamilton Therorem by computing $B^2-3B-2I$, where $B=\begin{bmatrix}3 & 1\\ -2 & 0\end{bmatrix}$

31. Verify Caley-Hamilton theorem and hence find the inverse of $A=\begin{bmatrix}1 & 4\\ 2 & 3\end{bmatrix}$ and hence evaluate A^4.

32. Find the characteristic equation of $A=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$ and then verify Caley Hamilton Theorem.

33. Use Cayley-Hamilton theorem for the matrix $A=\begin{bmatrix}1 & 4\\ 2 & 3\end{bmatrix}$ to express as a linear polynomial in A and evaluate $A^5-4A^4-7A^3+11A^2-A-10I$.

34. Using Caley Hamilton theorem find A^{-1} for $A=\begin{bmatrix}1 & -1\\ 2 & 3\end{bmatrix}$.

35. Find the eigenvalues and their multiplicity for $A=\begin{bmatrix}5 & -4 & 0\\ 1 & 0 & 2\\ 0 & 2 & 5\end{bmatrix}$

36. Find the algebraic multiplicity of each eigenvalue for the following:

a. A matrix A has eigenvalues 6,6,6,2

b. A matrix A has eigenvalues 5,5,1,3,6

37. Find eigenvalues of matrix $A=\begin{bmatrix}8 & -6 & 2\\ -6 & 7 & -4\\ 2 & -4 & 3\end{bmatrix}$.

38. For the given matrix $B=\begin{bmatrix}-2 & 2 & -3\\ 2 & 1 & -6\\ -1 & -2 & 0\end{bmatrix}$

a. Find characteristic equation of *B*.

b. Find eigen values and eigen vectors of matrix *B*.

39. Check that Algebraic Multiplicity (A.M.) is equal to Geometric Multiplicity (G.M.) holds

for the matrix $A=\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ -2 & -2 & -1 \end{bmatrix}$.

40. Find Algebraic Multiplicity and Geometric Multiplicity for the given matrix $A=\begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix}$

41. State whether the statement is true or False?

"For every eigenvalue of A, the geometric multiplicity is less than or equal to the algebraic multiplicity"

42. Find the geometric multiplicity of the eigen value $\lambda = 4$ for the matix $A=\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$

43. Find the algebraic multiplicity of the eigen value for the matrix $A=\begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$

44. For the characteristic equation of a matrix $(\lambda+3)^2(\lambda-5)=0$, find the algebraic multiplicity of each eigenvalue.

45. Can the geometric multiplicity of an eigenvalue exceed its algebraic multiplicity? Justify your answer by giving appropriate reason.

46. If a 3×3 matrix has three distinct eigenvalues, what can you say about its algebraic and geometric multiplicities?

47. Find the algebraic multiplicity of the eigenvalue for the matrix $B=\begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 0 \\ 0 & 1 & 2 \end{bmatrix}$.

48. Find the geometric multiplicity of the eigenvalue for the matrix $A=\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$.

49. If $D=\begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$ then find D^3.

50. If a matrix A is diagonalizable, what does this imply about its algebraic and geometric multiplicities?

51. If Geometric Multiplicity (G.M.) is less than Algebraic Multiplicity (A.M.) then what we can say about diagonalization?

52. Write only diagonal matrix D for a given matrix $A = \begin{bmatrix} 1 & 3 & 3 \\ 0 & 5 & -3 \\ 0 & 0 & 1 \end{bmatrix}$.

53. If $A = PDP^{-1}$ with the diagonal matrix is $D = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$. Find D^3.

54. For $A = \begin{bmatrix} 7 & 2 \\ -4 & 1 \end{bmatrix}$, find a fomula for A^k given that $A = PDP^{-1}$, where $P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$.

55. For a given diagonal matrix $A = \begin{bmatrix} -1 & 0 \\ 0 & 4 \end{bmatrix}$. Find A^5. Let $P = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ with $A = PDP^{-1}$ then find A^{10}

56. Prove that the given matrix $B = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 3 \\ 3 & 3 & 20 \end{bmatrix}$ is diagonalizable. Find the characteristic equation of the matrix.

57. Using encoding matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ encode and decode the message "ALL IS WELL".

58. Using encoding matrix $\begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ encode and decode the message "BE POSITIVE ALWAYS".

59. **Choose one correct option for the following:**

a. Trace of a matrix equals to ____.

{Product of eigenvalues, Sum of eigenvalues, Rank, Determinant}

b. For matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, the eigen vector for A is _____.

{(0,1), (1,0), (1,1), (2,0)}

c. If eigenvalues of a 2 X 2 matrix A are 2 and 3, then determinant value of A is ___.

{5,6,1,0}

d. For any system of Linear equations AX = B, if rank(A) = rank ([A | B]) = 2 and number of unknowns in system is equal to 2, then system is _____.

{consistent with a unique solution, consistent with infinite solution, inconsistent with no solution, with trivial solution only}

e. Algebraic multiplicity means _____.

{Number of eigenvectors repeated, Number of eigenvalues repeated, Rank of the matrix, Determinant of the matrix}

f. Solution of the system of Linear Equations: $x+y=2; x-y=0$ by using Gauss elimination method is _____.

{(1,1), (2,0), (0,2), (−1,1)}

g. If $A=\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$, then the sum of all the eigen values is ____.

{6, 9, 10, 4}

h. Principal Component Analysis is based on the concept of _____.

{Rank of the matrix, Eigenvalues of the matrix, Determinant value of the matrix LU Decomposition Method}

i. Using LU decomposition $A=\begin{bmatrix} 2 & 3 \\ 4 & 7 \end{bmatrix}$. Find U_____.

$\{U=\begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}, U=\begin{bmatrix} 2 & 3 \\ 0 & 7 \end{bmatrix}, U=\begin{bmatrix} 2 & 4 \\ 0 & 1 \end{bmatrix}, U=\begin{bmatrix} 2 & 4 \\ 0 & 7 \end{bmatrix}\}$

j. If $A=\begin{bmatrix} 2 & x-3 & x-2 \\ 3 & -2 & -1 \\ 4 & -1 & -5 \end{bmatrix}$ is a symmetric matrix then $x=$____.

{3, 6, 8, 0}

k. If $\begin{bmatrix} 2p+q & p-2q \\ 5r-s & 4r+3s \end{bmatrix}=\begin{bmatrix} 4 & -3 \\ 11 & 24 \end{bmatrix}$, then the value of $p+q-r+2s$ is_____.

{8, 10, 4, -8}

l. Which of the following is not a necessary condition for a matrix, say A, to be diagonalizable?

{A must have n linearly dependent eigen vectors, All the eigen values of A must be distinct, Arithmetic multiplicity of each eigen value must be equal to its Geometric multiplicity}

m. The eigen values of the matrix $A=\begin{bmatrix} 1 & 8 \\ 2 & 1 \end{bmatrix}$ are

{-3 and 5 , -3 and -5, 3 and 5, 3 and -5}

n. If $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$, then which one of the following equations is correct?

{ $A^3 - 3A^2 - 4A + 11I = 0$, $A^3 - 4A^2 - 3A + 11I = 0$

$A^3 + 4A^2 - 3A + 11I = 0$, $A^3 - 3A^2 + 4A + 11I = 0$ }

o. If $A = PDP^{-1}$, then A^k equals to

{ PDP^{-1}, PD^kP^{-1}, P^kDP^{-k}, ADP }

p. In the diagonalization process $A = PDP^{-1}$, what do the entries of the diagonal matrix D represents?

{The elements of the first row of matrix A, The eigenvalues of A , The components of the eigenvectors, The determinants of the submatrices}

17. A system of homogeneous linear equations $AX = 0$ is always _____.

{Inconsistent, Consistent, Having only one solution}

18. The trivial solution of the homogeneous system $AX = 0$ is obtained _____.

{if det(A) $\neq 0$, never , if det(A) = 0}

19. Which of the following is an orthogonal matrix?

{ $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, $\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ }

20. The rank of a matrix in row echelon form is equal to:

{the number of zero rows, the total number of rows, the number of non-zero rows, the number of columns}